NEURALPLANE: Structured 3D Reconstruction in Planar Primitives with Neural Fields

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Project page: https://neuralplane.github.io/

Abstract — Key Idea

- Reconstructs multi-view 3D plane maps with neural fields from inconsistent 2D plane observations.
- Distills geometric and semantic cues into a unified 3D neural representation.
- Annotation-free: leverages monocular priors and self-supervision.
- Outputs crisp planar primitives aligned with scene semantics.

Figure 1: Structured 3D reconstruction in planar primitives with neural fields (placeholder)

Introduction & Motivation

- Planar primitives are compact, expressive, and semantically meaningful for man-made scenes.
- Conventional geometry + RANSAC pipelines rely on noisy geometry and plane annotations.
- Requires joint modeling of geometry and semantics; handle cross-view inconsistencies.
- Neural fields enable fusing multi-view 2D cues into a consistent 3D representation.

Contributions

- NeuralPlane: multi-view 3D plane reconstruction without plane annotations.
- Self-prompting monocular module to excavate 2D plane regions and initialize local planar primitives.
- Plane-guided neural training with strong intra-primitive regularity for accurate geometry.
- Neural Coplanarity Field (NCF) and Neural Parser for semantic separation via contrastive learning.

Pipeline Overview

1) Monocular priors → SAM + normal clustering → 2D plane segments; estimate plane parameters; form Local Planar Primitives.

2) Plane-guided training → Geometry via density field with plane constraints; Semantics via NCF contrastive learning.

3) Extraction → Group by learned coplanarity features; fit planes (RANSAC); mesh triangulation.

Unified neural representation enables crisp, semantically coherent planar maps.

Fig. 1(a): Pipeline overview (placeholder)

Training Details

- Implicit geometry: volume density and view-dependent color (NeRF).
- Geometry branch: normal loss enforces consistency; pseudo-depth supervision with KL to rendering PDF; jointly refines plane parameters.
- Semantics branch: Neural Coplanarity Field with contrastive learning (pull within primitive, push across primitives gated by plane distance).
- Neural Parser: query-based prototypes supervised by DBSCAN centroids with Hungarian matching; learns instance separation during training.
- Overall loss: $L = L_{rgb} + \lambda_1 L_{normal} + \lambda_2 L_{p\text{-}depth} + \lambda_3 (L_{pull} + L_{push}) + L_{parser}$.

Generating Local Planar Primitives

- Plane segment M: consistent normals and geometric continuity.
- Normals from monocular predictor → K-means clustering; SAM auto masks with size thresholding.
- Initialize normal as average within segment; estimate offset via weighted distances to SfM keypoints (weights from reprojection error).

Eq: offset via minimizing weighted point-to-plane distances (placeholder)

Experimental Setup

- Datasets: 8 ScanNetv2 and 4 ScanNet++ indoor scenes.
- Baselines: PlanarRecon; geometry+RANSAC with MVS and neural implicit methods; AirPlanes.
- Metrics: Chamfer, F-score@5cm; RI, VOI, SC for semantics.
- Implementation: Nerfstudio/Nerfacto; dense SfM keypoints via COLMAP with LoFTR.

Results — Quantitative

- They attain state-of-the-art Chamfer, F-score, and segmentation metrics on ScanNetv2 and ScanNet++.
- AirPlanes achieves high precision but lower recall; the method balances both.

Table: ScanNetv2 (Chamfer↓, F-score↑, SC↑)

Quantitative table: ScanNetv2 Chamfer, F-score, SC across methods

Table: ScanNet++ (Chamfer↓, F-score↑, SC↑)

Quantitative table: ScanNet++ Chamfer, F-score, SC across methods

Summary chart: Chamfer↓ and F-score↑

Bar chart summarizing Chamfer and F-score across methods

Results — Qualitative

- Clean planar structures with fine details and coherent semantics on both datasets.

ScanNetv2 qualitative: input view, NeuralPlane, AirPlanes, PlanarRecon

Left to right: input view, NeuralPlane, AirPlanes, PlanarRecon (ScanNetv2)

ScanNet++ qualitative: input view, NeuralPlane, AirPlanes, PlanarRecon

Left to right: input view, NeuralPlane, AirPlanes, PlanarRecon (ScanNet++)

Implementation Notes

- Training uses 4k iterations with a batch of 8192 rays on a single NVIDIA RTX 3090.
- During the first 1k iterations, L_p-depth and L_push are disabled for primitives with fewer than 50 keypoints.
- Offsets are then re-estimated via rendered depth; the full loss is enabled to refine local geometry.
- Key hyperparameters: feature dim d=4; thresholds (t_o, t_n)=(8 cm, cos 10°); number of prototypes N_p=32.

Discussion & Takeaways

- Joint geometry–semantics learning resolves coplanar-but-distinct instances.
- Annotation-free plane cues suffice when distilled via neural fields.
- Fast training with adjustable level of detail via prototype count.
- The NeuralPlane@PlaneRecTR variant validates the monocular primitive generation.

Limitations

- Relies on monocular normal priors and SAM over-segmentation; severe low-texture scenes can yield sparse/misaligned SfM keypoints affecting initialization.
- Over-segmentation introduces cross-view inconsistencies that require sufficient multi-view coverage to reconcile.
- NCF contrastive learning may under-separate large coplanar regions with subtle semantic differences when pushing thresholds are conservative.
- Two-stage extraction (feature grouping → RANSAC → meshing) can miss very small planar pieces.

Open Questions

- How can inconsistent 2D plane observations be fused with minimal reliance on SfM keypoints in low-texture scenes?
- Can adaptive semantics-aware pushing thresholds improve separation of subtly different coplanar instances?
- How can level-of-detail control be integrated directly into training to automatically match downstream requirements?

Conclusion

- Reconstructs high-fidelity planar primitives aligned with semantics.
- The method combines monocular priors, plane-guided training, NCF, and Neural Parser.
- They deliver state-of-the-art results on ScanNetv2 and ScanNet++ with efficient training.